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# Unit -1

## Introduction to Python and Core Basics

1. Write a python program to print "Hello World"
2. Write a python program to get user input using input( ) function
3. WAP to take only numerical input
4. Convert the number to floating point number
5. WAP to Add, Subtract, Multiply and Divide 2 numbers
6. Print the quotient and remainder separately for division operation
7. Write a program to find meter to kilometer.
8. Write a program to find area of a rectangle. (Area=l*b). Take input parameters from user
9. Write a program to find volume of cube. (Area=l*b*h). Take input parameters from user
10. Write a program to find area of triangle. (Area=(l*b)/2). Take input parameters from user
11. WAP to convert the given temperature from Fahrenheit to Celsius using the formula C = (F – 32) / 1.8
12. WAP to convert temperature from Celsius to Fahrenheit where temperature in Celsius is entered by user.
    (F= C*(9/5) + 32)
13. Take a number as user input and convert it into Binary, Octal and Hexadecimal numbers

14. Take binary, octal and hexadecimal numbers as an input and convert them to Decimal number
15. Take 3 different inputs from user and display their data type.
16. Perform binary “AND” and “OR” operation for given 2 integer numbers from user input
17. Write a program to calculate area of circle. (pi*r*r)
18. Write a program to calculate simple interest using formula SI = (P*R*N) / 100. Take all parameters as input from user
19. Without applying condition statement display output as “true” if a number is an even number and “false” if the number is an odd number
20. Write a program to swap two numbers using a third variable and also without using the third variable.
21. Write a program to display the operations of bitwise AND and bitwise OR on two numbers entered by the user
22. Find the output for the following code
23. X=100

    Y=200

    Print(x and y)

    Print (x or y). Update the values of x and y to 0 and then check the output again. Analyse the result in regards to python

## UNIT -2

## Program Control Flow and Python Data Structures

1. Find the minimum and maximum of 2 numbers
2. Write a program to find the maximum of three numbers
3. Write a program to input the basic salary and calculate the net salary based on the given conditions. If basic is <10000 then da=25%, hra=5%. If basic>=10000 and basic <=30000 then da=35%, hra=10%. If basic >30000 then da=40%, hra=20%. Pf is same for all 12%.
4. Print 1 to 10 numbers in ascending and descending order using range
5. Print odd numbers between 1 to 50
6. Print the ‘*’ patterns using range()

7. Print the number pyramid using range()
8. Print all prime numbers between 10 to 50
9. Take 10 numbers (min 3 digits) as user input and check whether a number is palindrome or not.
10. WAP to check whether the given number is Armstrong or not.
11. WAP to input n numbers and count the total number of odd and even numbers in the tuple.
12. Print multiple lines using single print statement. as –

    I like "Python Programming" very much
    It is my favorite subject
13. Print a part of the above string "very much" using the slice operator.
14. Print the last 5 characters from the above given string
15. Print only the second line of the given string
16. Take two strings as input from the user and concatenate them.
17. Take a number and a string from the user and repeat the string for that many times.
18. Take an input character from the user and check whether that character is present in the above given string or not. – Using 'in' operator and using 'not in' operator
19. Create a menu driven program for string manipulation
    a. Find the length of a string
    b. Print the string in upper case
    c. Print the string in lower case
    d. Print the string with initial capital
    e. Split the string based on the character entered.
20. Take two strings as input s1 and s2 and check whether s2 is present in s1 or not.
21. If s2 is a part of s1 then print the 1st and last occurrences of it
22. If s2 is present in s1 then also count number of times it occurs in s1.
23. Count total number of words in the string input by user
24. Take a string input from user and print it in reverse using range
25. 
26. Take a tuple input and print all the elements of it in reverse sequence
27. Copy the inputted string to another string by replacing the character 'o' with '@' Eg. 'Hello' will be copied to another string as 'Hell@' and 'Good Morning' will become 'G@@d M@rning' (Without using replace())
28. Take a string as an input from the user. Find total number of vowels in it.

(Hint: take a tuple of vowels)

29. Create a tuple for name say t1 (FirstName, MiddleName, LastName)
30. Create a tuple say t2 for marks of 5 subjects
31. Make a total of all the marks and print it. (with and without using sum() method)
32. Make a tuple t3 having 2 elements as t1 and t2 (tuples created above) – It is called a nested tuple
33. Take an input number and find whether that is present as an element in the tuple t3 or not.
34. Create a tuple of 5 fruits. Ask the user to input a fruit name and search that name in the given fruit tuple. Display suitable messages
35. Create a tuple of cities of Gujarat by taking user input.
36. Find the length of name of each city in the above tuple. With and without len() method
37. Create a nested tuple t4 of your (name, (hobbies), (friends), degree)
38. Find an element in the nested tuple (t4) and print its position if found, otherwise print "Not found"
39. Take a tuple of 10 integer numbers and segregate odd and even numbers in 2 different tuples.
40. Create a list of students say L1
    a. Count total number of students from the list L1
    b. Add one more student in the list L1
    c. Display all the students in the sorted order
    d. Check a particular student's name is present in the list or not
    e. If the student's name is present in the list, print total number of same name students in the list L1 and display the position of 1st student
    f. Remove the last student from the list L1
    g. Remove a particular student from the list. (Take a name of student from the user.)
    h. While removing the student from the list, if multiple students have same name then remove all of them from the list.
41. Create a list of 10 numbers and find the maximum and minimum numbers from it.
42. Create a list of alphabets and count total number of vowels in it.
43. Create a list of even numbers between 1 to 21 using range()
44. Create a list of employees (nested list) with their personal details like [name, age, salary, expertise] in a list. Ask the user to enter name and display the

details of that employee. If the employee is not in the list, print error message.

45. Create a list by taking any 5 inputs from the user.
46. Display the students from L1 list, whose name contains the character ‘a’.
47. Create a list of 10 numbers and find the total of odd numbers and even numbers
48. Put all the odd numbers in one list and even numbers in another list.
49. Create a list having mixed elements like string and integers. Print only integer elements from the list with and without using list comprehension
50. Write a Python program to store student roll numbers in a set and prepare the sets according to the requirement of the question
    a. Write a program to add new students to an existing class attendance set.
    b. Write a program to remove students who are absent from the attendance set.
51. Create a sets of Courses and write a program to check whether a given student is enrolled in a particular course or not.
    a. Write a program to find students who are enrolled in both the courses.
52. Create two sets of elective subjects chosen by the students. Write a program to find students who are enrolled in at least one of two elective subjects
    a. Write a program to find students who are enrolled only in Course A and not in Course B.
    b. Write a program to find students who participated in exactly one of two courses.
    c. Write a program to remove duplicate student from course sets.
53. Given the set
    students = {"Amit", "Neha", "Riya", "Karan"}
    write a Python program to check whether "Riya" is enrolled in the course.
54. Given the sets
    math_students = {"Amit", "Neha", "Riya"}
    cs_students = {"Riya", "Karan", "Pooja"}
    write a program to find students enrolled in **both** subjects.
55. Given the sets
    club_A = {"Rahul", "Sneha", "Amit"}
    club_B = {"Sneha", "Karan", "Pooja"}
    write a program to find students who are members of **at least one club**.

56. Given the sets
course_A = {"Amit", "Neha", "Riya", "Karan"}
course_B = {"Neha", "Karan"}
write a program to find students enrolled **only in Course A**.
57. Given the sets
workshop1 = {"Amit", "Riya", "Pooja"}
workshop2 = {"Riya", "Karan", "Neha"}
write a program to find students who attended **exactly one workshop**.
58. Given the set
attendance = {"Amit", "Neha", "Riya", "Karan"}
write a program to remove "Neha" from the attendance list.
59. Given the set
present_students = {"Ravi", "Sneha", "Amit"}
write a program to display all students using a loop.
60. Given the list
emails = ["a@gmail.com", "b@gmail.com", "a@gmail.com", "c@gmail.com"]
write a program to remove duplicate email IDs using a set.
61. Given the sets
class_A = {"Amit", "Neha"}
class_B = {"Amit", "Neha", "Riya", "Karan"}
write a program to check whether **Class A is a subset of Class B**.
62. Given the sets
team1 = {"Amit", "Riya"}
team2 = {"Karan", "Neha"}
write a program to check whether the two teams are **disjoint**.

63. Create a dictionary of employees where empId will be the key and value will be the name of an employee
    1. Display how many employees are there in the dictionary.
    2. Display all empID and add them in a separate list.
    3. Display all employee names and take them to a separate list
    4. Take an empId from the user and check if that employee is there in the dictionary or not.
    5. If an empID is there in the dictionary then display the name of

that employee or if not available then add an ID and Name of the employee in the dictionary
6. Change the name of the employee of empID taken by the user
7. Remove an employee whose ID is provided by the user

64. Take 5 names of students as an input from the user and create a dictionary with keys as their initials and value is a list as [age, degree, favorite subject]
    1. Display the youngest student from the above dictionary.
    2. Create a dictionary of students having rollno of the student is as key and value is a list of marks obtained by that student in 5 subjects
    3. Create a dictionary from the above one, where key is rollno and value is (total of all subjects, percentage and grade ) a tuple of his result
    4. Display the rollno who has scored highest marks (total)
    5. Take 10 numbers from the user and create a list, apply bubble sort and arrange the elements in the list.

## UNIT -3
## Functions and Files in Python

1. Take user input and create a menu driven program to perform mathematical operations like addition, subtraction, multiplication, division, integer division, power. Return values from the functions
2. Create functions to calculate
    a. Area of a rectangle = width * length
    b. Area of a triangle = ½ * Height * Base
    c. Area of a circle = pi*r*r
3. Create functions to convert decimal numbers to binary, octal and hexadecimal numbers. Always return values from the functions
4. Write an UDF to return a list having only unique values by removing duplicate values from the provided input list.

Eg. Sample List : [1,2,3,3,3,3,4,5]

Unique List : [1, 2, 3, 4, 5]

5. Write a Python function to multiply all the numbers in a list.
6. Write a UDF to check the inputted number is between specified range or not.
7. Write a function to calculate total number of Uppercase and lowercase characters in the string.
8. Write an UDF to check if the user given number is a prime number or not.
9. Write a findString() function to find all the positions of occurrences of string2 in string1 and return that value. If string2 is not present in string1 then display suitable message.

   Eg. Str1 = Hello all, Good Morning to all. (pass it as a parameter in the function)

   Str2 = 'all' (pass it as a parameter, but f not passed take a default argument)

   O/p: String 2 found at positions: [6, 27]

10. Create a list of fruits and using different functions perform the following operations: Show the use of globals() and don't return from the functions
    a. Add a fruit at the last
    b. Insert a fruit at a particular position (pass it as an argument. If the position is not passed then take default argument as 1)
    c. Update the fruit (use keyword arguments)
    d. Remove a fruit from the list (pass an index position/ pass a name of the fruit as an argument)
    e. Arrange the fruits in an order

11 Create a function to generate prime numbers. Ask total numbers form the user and pass in the function which will return a list of prime numbers.

12 Eg. GeneratePrime(10) function will return $1^{st}$ 10 prime numbers starting from 2 like 2,3,5,7,11,13,15,1719,23

13 Create a lambda function that will return maximum of two numbers

14 Create a lambda function that will return maximum of three numbers

15 Write a lambda function that takes one number and if the number is even, returns that number multiplied by 5 else if the number is odd, returns that number multiplied by 10
16 Take a list of mixed elements and
    a. Write a lambda function to separate integer elements as an output list.
    b. Write another lambda function to separate string elements as an output list.
17 Modify the above program using filter()
18 Filter all vowels from the given string.
19 From the provided list filter, the even numbers and odd numbers as a separate output list
20 Write a lambda function that will take 2 inputs. If inputs are integers, it will return the product of 2 numbers. Else perform concatenation.
21 Sort the list elements using lambda
22 Find the average of all the elements passed as an argument in lambda (using variable length arguments)
23 Find the square of each element of a list (using map())
24 Use a lambda function to calculate grades for a list of scores (using map())
Eg scores = [88, 92, 78, 95, 86]
25 Add all the elements of the list (using reduce())
26 Multiply all the elements of the list (using reduce())
27 Find the maximum element from the list using reduce()
28 Sorting the dictionary elements using lambda (by using sorted () method) according to age and if age is same then sort my name
Eg. stud= [{'name': 'Amit', 'age': 25}, {'name': 'Bina', 'age': 22}, {'name': 'Dax', 'age': 25}]
29 Take 2 lists and add the elements of it if the 1st number is greater than the other else find the difference between them
Eg. nums1 = [6, 5, 3, 9]   nums2 = [0, 1, 7, 7]
O/P [6, 4, 10, 2]
30 Take a list of person names and display them all in upper case using map()
31 Take a list of floating-point numbers and display list of all round numbers. Also round them with just 2 decimal points. Using map()
Eg. [6.56773, 9.57668, 4.00914, 56.24241, 9.01344]

o/p [7, 10, 4, 56, 9] and [6.57, 9.58, 4.01, 56.24, 9.01]

32 Take a list of words and print all palindrome words using filter() [Hint: string slicing str1[::-1]]

33 Take a list of students and filter the students whose name is less than 6 characters.

34 Take a string as an input and display the output to analysis the string based on separate words. Using map()

   a. Display the words in upper case along with the length of each word
   b. Display total number of each vowel in each word

Eg. Str1 = 'Hello how are you?'

o/p: [{'a': 0, 'e': 1, 'length': 5}, {'a': 0, 'e': 0, 'length': 3}, {'a': 1, 'e': 1, 'length': 3}, {'a': 0, 'e': 0, 'length': 4}]

35 Take a matrix as input and transpose its elements using lambda

Eg. matrix = [[1, 2],[3,4],[5,6],[7,8]]

o/p: [[1, 3, 5, 7], [2, 4, 6, 8]]

36 Find the factorial of a number using lambda (recursive)

37 Create a menu driven program with user defined functions to insert update delete elements in the dictionary object of employees

Emp = {empCode:[name, age, salary, (expert areas)],…..}

38 Write a python script to generate result for a particular student.

1. Create a student data in a dictionary object as shown below:

```
stud = {1: {"name":'Amit', "age":23,"marks": [(10,15,12), (11,12,13)]},
        2: {"name":'Bhumi',"age":22,"marks": [(13,15,11), (10,10,13)]},
        3: {"name":'Bharat',"age":23,"marks": [(12,12,14), (13,14,15)]}
  }
```

NOTE: Here students are getting marks of 3 subjects in 2 attempts of test in a form of tuple

2. Create separate user defined functions to (Create a menu for options)

   a. Take user input for creating entry of any new student (addStud())

b. Print all marks of a specific student. (Result(name))

c. Display overall result of all students in a given format

39 Create a text file with different modes like w, w+, a, a+ and write few lines in it

40 Read the content of the whole file together

41 Print the length of the file data

42 Read the file content line by line

43 Print total number of words in each line in the file

44 Print all the words in reverse.

45 Write multiple lines in a text file. Using list object

46 Take a filename from the user to read that file

47 If the file to be read is not available then print suitable message

48 After reading the file content, append the text at the end of the file.

49 Open a file and append a line at the beginning of the file content

50 Copy the content of one file to another

51 Read $10^{th}$ to $15^{th}$ byte from the file and print.

52 Read an existing file and take a user input string to be appended in that file. Also ask the position where new line need to be appended. Update the file content and print the updated file. [Hint: Make a file with new line character after each line]

53 Read an alternate bytes/ character from the file.

54 Read alternate lines from the file.

1. Write a python script to read the text file content and print the output in form of line wise total words in the file.
   <u>File Content as below:</u>
   Hello, How are you?
   Very good morning
   Have a nice day to all
   Good Bye…
   Output: [(1,4), (2,3), (3,6), (4,2)]
2. Open a text file using with statement and write and read the content from that file.
3. Take the user input for data to be written in the text file. Enter the data line

by line, till '@' character is entered by the user at the end.

4. Create a text file having string and numeric data. Write a script to separate the string and numbers in two different files [Hint: ch.isdigit() method will return true , if character is a number]
5. Create a menu driven program to perform various file operations through python functions as:
   a) Create a file – (define the filename as a default argument)
   b) Read the content of a specified file – Return the content in a string
   c) Append the content in the specified file
   d) Rename a file – (Take filename as keyword arguments)
   e) Delete a file - (define the filename as a default argument)
   f) Create a directory / folder
   g) Display all the files present in the specified folder
   h) Display only .txt file names from the specified folder
   i) Display the files, starting with letter 't' in their filename.
   j) Display all python files from a specified folder.(Either .py extension or filename/ folder name contains 'py' in between)
   k) Display the file names having .txt extension

## UNIT -4
## Exception Handling Database Connectivity

1. Write a Python script to take a user input to enter a list of elements for employee data and write it into a file. Handle the following exceptions for it.
   a. If an age entered is not a number
   b. If salary is not defined and trying to append in the list
   c. Entered age must be between 18 and 25 only
   d. Salary must be greater than or equal to 10000
   e. Calculate HRA and check for the ZeroDivisionError
   f. Open a file in read mode and try to write into it
   g. Try to write the whole list object into a file

**NOTE:**

- **TypeError**: This exception is raised when an operation or function is applied to an object of the wrong type, such as adding a string to an integer.

- **NameError**: This exception is raised when a variable or function name is not found in the current scope.
- **IndexError**: This exception is raised when an index is out of range for a list, tuple, or other sequence types.
- **KeyError**: This exception is raised when a key is not found in a dictionary.
- **ValueError**: This exception is raised when a function or method is called with an invalid argument or input, such as trying to convert a string to an integer when the string does not represent a valid integer.
- **AttributeError**: This exception is raised when an attribute or method is not found on an object, such as trying to access a non-existent attribute of a class instance.
- **IOError**: This exception is raised when an I/O operation, such as reading or writing a file, fails due to an input/output error.
- **ZeroDivisionError**: This exception is raised when an attempt is made to divide a number by zero.

**ImportError**: This exception is raised when an import statement fails to find or load a module.

2. Create a dictionary object for student details (Rollno, name, age, hobby, marks…)
3. Create a user defined exception if entered marks are > 50
4. Store all those dictionary data in a binary file.
5. Create separate functions for addData, updateData, deleteData from the binary file

**Essential Assignment:**

1. Create a Database in MySQL – name MyDB
2. Create a table – employee having fields – eno, ename, age, salary, doj
3. Insert, update and delete specific rows in that table using python script
4. Create a menu driven CRUD operation program to perform above tasks.

Help:
mysql.connector module
db = connector.connect(host="localhost",user="root",password="",
database="myDB")
cur = db.cursor()
cur.execute()

```
row = cur.fetchall()
cur.executemany()
db.commit()
```

5. Create a MySQL database table called Tournament. Use exception handling while creating a database table.
6. Insert the records from the readymade text file.
7. Use exception handling while opening and reading the text file.
8. Separate Text files for each sport contains Players details like (name, age, no_of_Tournaments_Played)
9. There should be only one DB table for all sports players, having fields like Name, Age, Sport_Play, Tournaments
10. After inserting data in the table, write a menu driven program to
    a. Search sport wise no. of players, Tournament wise players list
    b. Update the tournament field (increment by 1) for a user specified player (when he has played match)
11. Take the DB table backup in the binary file. (Use Pickle.dump())

## Unit -5

## GUI , Data Analysis with Pandas & Visualization with matplotlib

1. Create a GUI program that takes user input in the Entry widget and on button click that text must be displayed on the Label widget
    1. Insert a record in the DB table by taking user input from the GUI
2. Update and delete the record in the DB table.
3. Create a Calculator application using tkinter module in Python
4. Perform different mathematical operations using GUI – button widget
5. Create a GUI for File handling operations like:
    a. Open a text file – the name provided by the user
    b. Create a new text file

   c. Edit the content of the text file
   d. Delete the specified text file
   e. Search a specific word in the text file.
6. Create a GUI based program to take a number from the user and separate that number in 2 different files based on if the number is a prime number or not.
7. Create a dataframe from a .csv file
8. Create a dataframe from the dictionary object
9. Create a dataframe from a list object
10. Create a dataframe from an excel sheet. (Student data: RollNo, Name, Age, Marks)

   1. Display total number of rows and columns in the dataframe
   2. Display only 1st 3 rows from dataframe
   3. Display only last two rows from dataframe
   4. Display 3rd to 7th row of the dsataframe
   5. Display all the rows in reverse order.
   6. Display all column names of the dataframe.
   7. Display only name and age of all students from the dataframe
   8. Display maximum and minimum marks from the dataframe.
   9. Display the statistical analysis of marks from the student dataframe
   10. Display the name of the student having marks > 50
   11. Display the rollno and name of the student whose age is > 20
   12. Display the students having age between 20 and 25
   13. Display the name of the student who has scored maximum marks
   14. Display the students who have scored more than average marks (use mean)
   15. Change the index in DataFrame and create a new Dataframe
   16. Modify the original DataFrame by changing the Index inplace.
   17. Search for a particular row using index value

18. Reset the index
19. Arrange all the students in alphabetical order of their names
20. Arrange all the students according to their marks in descending order
21. Display the missing marks with 0
22. Display only those students who have scored more than 0. (drop the missing value row)
23. Display the DataFrame with suitable message for missing data

11. Create a scalar series (dictionary object with single value) and convert it into dataframe
12. Create MultiIndex.from_arrays like Students [ ], Score [ ], Age [ ]
13. Create MultiIndex.from_frame –
    a. Use dictionary object for employee data for empId, Name and Salary
    b. Create DataFrames by read_excel(), read_csv() and set multiindex
14. Create DataFrame from List object with Index values
    List object contains details for 5 students' result: Pass /Fail with scores in 3 subjects
15. Find total number of 'pass' students and 'fail' students in each subject from above list – use of groupby
16. Find minimum and maximum marks of each subject

17. Create 2 data frames from SQL tables and merge them based on common column (keys)
18. Plot a line graph,
19. Make a letter 'A' using plot method with x and y points and then after passing the array of points (multiple lines)
20. Make letters 'E', 'F', 'H', 'I', 'K', 'L', 'M', 'W', 'X' and 'Z' using plot()
21. Use of linestyle as 'dotted' , 'dashed', 'dashdot' for plotting above characters, Make all letters colourful, set different line width, use different markers
22. Make a diamond shape using multiple lnes
23. Set the graph title, xlabel and ylabel
24. Draw grids with (x and y axis , linestyle, linewidth)
25. Save all line graphs in separate .png files
26. Draw bargraph for selling of different electronic gadgets
27. Draw a histogram for the yearly performance of a student
28. Draw a pie chart for the student's participation in different games

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